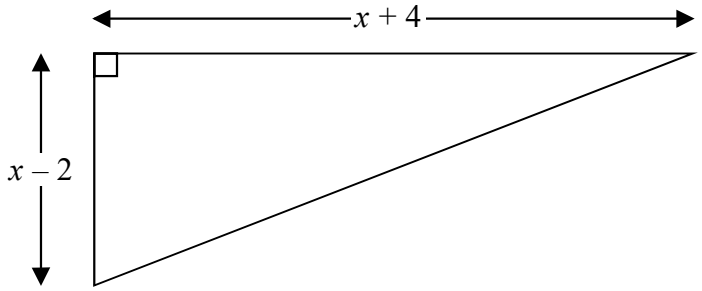
14 The diagram shows a right-angled triangle.



All the measurements are in centimetres.

The area of the triangle is 27.5 cm^2

Work out the length of the shortest side of the triangle.
You must show all your working.

½(x – 2)(x + 4) = 27.5

(x – 2)(x + 4) = 55

x^2 + 2x – 63 = 0

(x – 7)(x + 9) = 0

x = 7 (a positive side); shortest = x – 2 = 5 cm.

4 marks total
Set area equation
Expand to quadratic
Factor and solve x = 7
Shortest side 5 cm

..... cm

(Total for Question 14 is 4 marks)

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